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PACE: Pareto-Adaptive Compression for Efficient Native MLLMs

Orthogonal Spatial Saliency Decomposition Achieves Near-Lossless Performance at 25–50% Token Retention with 2× FLOPs Reduction

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Native multimodal large language models process image patches directly, without a separate vision encoder, but this architectural simplicity comes at a cost: no encoder-derived saliency signal is available for token compression. PACE addresses this gap with an orthogonal spatial attention decomposition and a Pareto-optimal budget allocation head. At 50% visual token retention, PACE delivers near-lossless performance on standard benchmarks and more than 2× FLOPs reduction; at 25% retention it improves on chart-understanding tasks while remaining competitive elsewhere. Accepted at NeurIPS 2026.

AT A GLANCE

Problem: ViT-free (native) MLLMs have no encoder attention signal to guide token compression; existing methods are inapplicable.

Why it matters: Native MLLMs are increasingly popular; without compression, visual token sequences create prohibitive attention cost.

Method: Decompose attention into orthogonal height and width saliency components gated by a positional signal; allocate token budgets via Pareto optimization over performance, efficiency, and spatial coverage.

Result: Near-lossless accuracy at 50% retention; >2× FLOPs reduction; improvement on chart benchmarks at 25% retention.

Evidence: Evaluation across multiple native MLLM architectures and benchmarks; ablations confirm each component's contribution.

The Big Picture

The field has converged on two approaches to multimodal LLMs. The dominant approach uses a separate vision encoder (ViT) to process images into a compact visual representation before passing it to the language model. The emerging alternative—the *native* or *ViT-free* MLLM—processes raw image patches directly in the transformer, unifying modalities from the start.

Native architectures reduce engineering complexity and enable end-to-end optimization, but they reintroduce a well-known problem at scale: the number of visual tokens grows quadratically with image resolution. For a 448×448 image at a patch size of 14, that is 1024 tokens—before any text. Attention cost is $O(N^2)$; this is already prohibitive at high resolution or in multi-image settings.

Compressing visual tokens is the natural remedy, but every existing compression method was designed for ViT-based pipelines, where the encoder's [CLS] token attention or patch-similarity matrix provides a natural saliency signal. Native models have neither. PACE is the first method designed specifically for this setting.

Why Existing Approaches Fall Short

ViT-based compression (ToMe, FastViT, DynamicViT): All rely on the encoder's attention map or token similarity computed within the ViT. These signals do not exist in native architectures; applying these methods to native models produces near-random compression.

Uniform subsampling: Discards spatial structure uniformly, missing locally salient regions that are scattered across the image.

Greedy attention thresholding: Retains the tokens with the highest self-attention scores in the LLM; this greedy approach ignores spatial coverage and can cluster selections around a single prominent region, missing the rest of the image.

Core Mechanism

PACE introduces two components applied sequentially at each layer of the native MLLM.

Orthogonal spatial saliency decomposition: The attention matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ for the visual tokens is factored into marginals along the height and width dimensions of the image grid:

$$\mathbf{A} \approx \mathbf{A}_H \otimes \mathbf{A}_W$$

where $\mathbf{A}_H \in \mathbb{R}^{h \times h}$ and $\mathbf{A}_W \in \mathbb{R}^{w \times w}$ capture row-wise and column-wise saliency, with $N = h \times w$. A positional attention signal p_{ij} (a learned embedding of the token's position relative to attended regions) gates the combined score.

Pareto budget allocation: Rather than fixing a single compression ratio for all inputs, PACE frames token budget selection as a Pareto optimization over three objectives: performance preservation, FLOPs reduction, and spatial coverage. A lightweight allocation network selects the operating point at inference time.

Technical Formulation

The saliency score for token (i, j) is:

$$s_{ij} = \alpha\,[\mathbf{A}_H]_{ii} + \beta\,[\mathbf{A}_W]_{jj} + \gamma\,p_{ij}$$

where (α, β, γ) are learned scalars. The top-B tokens by score are retained, where B is determined by the allocation network.

The Pareto budget allocation problem is:

$$\min_{\mathbf{m}} \ (\text{PerfDrop}(\mathbf{m}),\ -\text{FLOPs_red}(\mathbf{m}),\ -\text{SpatCov}(\mathbf{m}))$$

subject to $\|\mathbf{m}\|_0 = B$, where $\mathbf{m}$ is the binary selection mask. The Pareto frontier is computed on a held-out calibration set and parameterised by a small MLP that maps input complexity features (image entropy, text query length) to a budget B and weighting (α, β, γ).

At training time, the allocation MLP is trained to minimise the weighted Chebyshev distance to the Pareto frontier:

$$\mathcal{L}_{\text{Pareto}}(\phi) = \mathbb{E}[\max(\lambda_1\,\Delta\text{Perf},\ \lambda_2\,\Delta\text{FLOPs},\ \lambda_3\,\Delta\text{Cov})]$$

Learning or Inference Procedure

1. Construct the Pareto frontier by running the model at several compression ratios on a calibration set; record performance, FLOPs, and spatial coverage.
2. Train the allocation MLP on the frontier with $\mathcal{L}_{\text{Pareto}}$.
3. At inference: the allocation MLP selects budget B and weightings; saliency scores are computed; the top-B tokens are retained; the remaining tokens are pruned before subsequent transformer layers.
4. PACE is a plug-in module; the base MLLM is not modified or fine-tuned.

What the Guarantee Says

The Pareto formulation provides a formal guarantee: the selected budget B lies on the Pareto frontier (no other $B' \neq B$ strictly improves all three objectives simultaneously). The spatial coverage constraint ensures that selection never degenerates into clustering around a single image region, bounding the worst-case missed-region probability.

Experimental Findings

- **At 50% token retention:** Near-lossless accuracy on VQA, spatial reasoning, and multimodal reasoning benchmarks; FLOPs reduced by 2.1–2.4×.
- **At 25% token retention:** Performance improves on chart understanding (spatial coverage prevents missing chart regions that a greedy method would skip); slight degradation on spatial reasoning.
- **Cross-architecture:** Gains are consistent across multiple native MLLM architectures, confirming that PACE is not tuned to a single model's attention pattern.
- **Positional signal ablation:** Removing the positional gate p_{ij} hurts spatial-reasoning tasks the most, where token position encodes geometric relationships critical for the answer.

Ablations and Interpretation

Without Pareto allocation (fixed global ratio): Uniform 50% budget performs well on average but degrades on information-dense inputs (too aggressive) and wastes compute on information-sparse inputs (not aggressive enough).

Without orthogonal decomposition (scalar per-token attention): The saliency map is blurry; clusters of redundant background tokens score similarly to salient foreground tokens.

Height-only or width-only saliency: Misses salient regions localised in the orthogonal direction (e.g., a horizontal strip of text is missed by height-only saliency when the strip occupies few rows).

Why native MLLMs tolerate compression: The authors hypothesise that without ViT pre-processing, the transformer distributes attention more uniformly in early layers and learns implicit token-aggregation in later layers; explicit compression makes this implicit process explicit.

Reference

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